

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1501
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I A.Reddy Prasad (192472045) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Reddy Prasad

Assessment Tasks Industry Problem Solving Task

Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Social Media Fraud Detection Problem - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Government Data Platform Problem A government agency is building a national data platform.

It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

Q1-Retail Data Optimization

A retail company generates large volumes of **sales, inventory, and customer behavior data** from physical stores and its online platform. A scalable Big Data architecture can integrate these different data sources and process them in real time.

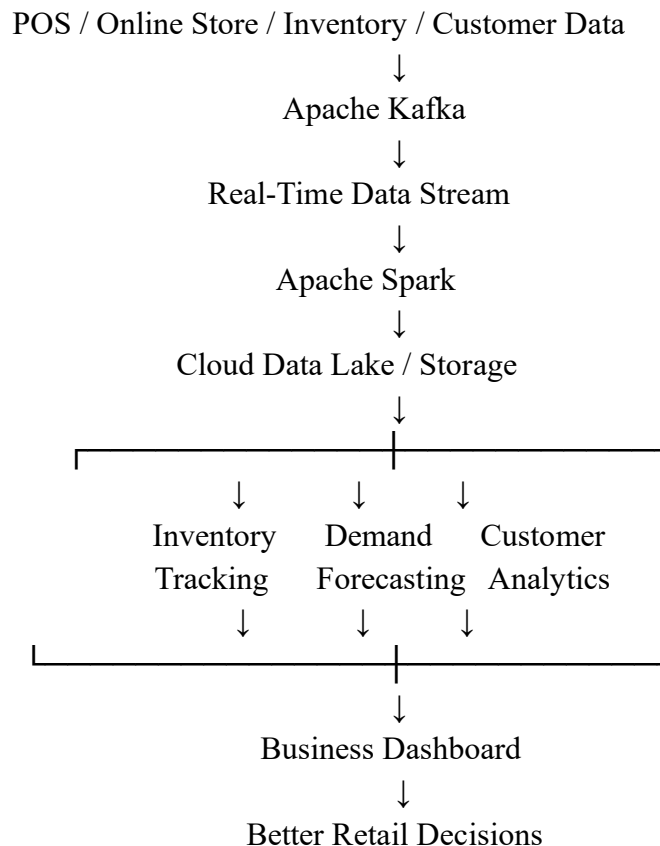
The proposed architecture collects data from **POS systems, e-commerce transactions, inventory systems, customer interactions, and website/mobile applications**. The data can be stored in a scalable cloud data lake or distributed storage system. **Apache Kafka** can be used for real-time data streaming, while **Apache Spark/PySpark** can process both real-time and historical data.

For **real-time inventory tracking**, streaming data from stores and online orders is processed continuously to update stock levels. This helps the company identify low-stock products and avoid overstocking or stock-outs.

For **demand forecasting**, historical sales data can be analyzed using machine learning models such as **Linear Regression, Random Forest, or time-series forecasting** to predict future product demand. The predicted demand can help the company plan inventory and restocking.

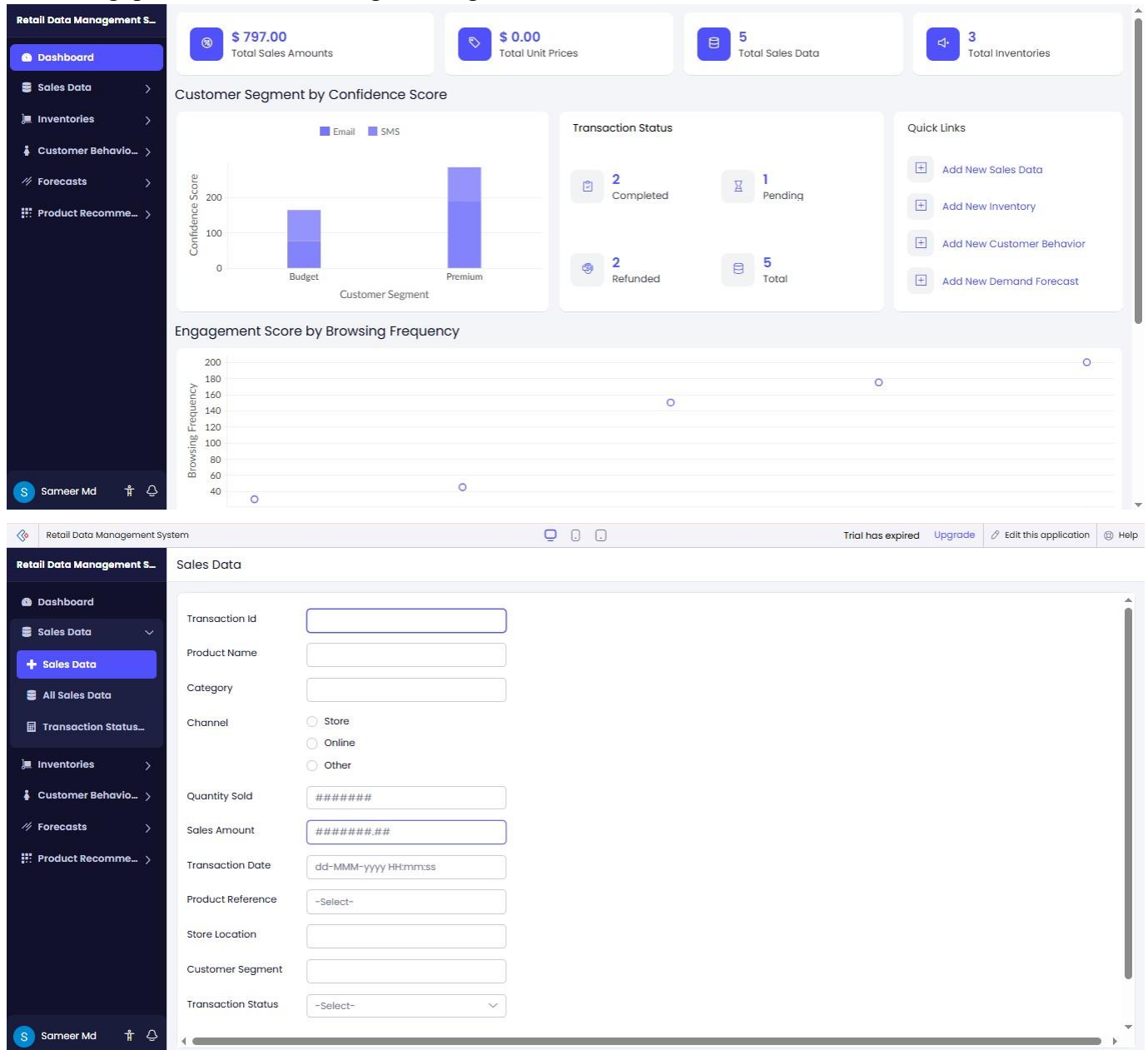
For **personalized marketing**, customer behavior such as purchases, searches, and product interactions can be analyzed using **recommendation and customer-segmentation techniques**. For example, a recommendation model can suggest products based on a customer's previous purchases and browsing behavior.

Architecture



This architecture is **scalable** because distributed processing can handle increasing volumes of retail data. It also supports real-time decision-making and can be expanded as the number of stores, customers, and transactions increases.

For security, the company should implement **authentication, role-based access control, encryption, secure data transmission, and monitoring** to protect customer and business data. Overall, the Big Data solution helps the retailer maintain accurate inventory, forecast demand, understand customer behavior, and provide personalized recommendations. This improves operational efficiency, reduces stock-related problems, and supports better customer service. The solution directly addresses the assessment requirements for real-time inventory tracking, demand forecasting, personalized marketing, and Big Data architecture.



Retail Data Management S... Dashboard Sales Data + Sales Data All Sales Data Transaction Status... Inventories Customer Behavior... Forecasts Product Recomm...	All Sales Data		Edit Duplicate More X
	Transaction Id	Product Name	Overview
	T005	Smart Watch	Transaction Id
	T004	Yoga Mat	Product Name
	T003	Running Shoes	Category
T002 Organic Coffee Beans T001 Wireless Headphones			Channel
			Quantity Sold
			Sales Amount
			Transaction Date
			Product Reference
			Store Location
			Customer Segment
			Transaction Status
Showing 5 of 5			Product Reference
creatoran.zoh.in/saveethachoolinfengineerinn/retail-data-management-custem			No records found

Q2 – Social Media Fraud Detection

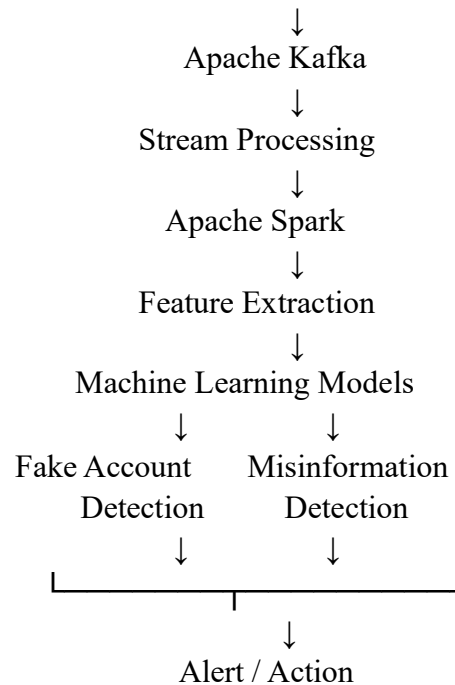
A scalable Big Data fraud detection system can collect data from **posts, comments, likes, shares, user profiles, login activities, and interaction patterns**. Since the data is generated continuously, a stream-processing architecture can be used for near-real-time analysis. **Apache Kafka** can collect and stream user interactions from different sources. **Apache Spark Streaming/PySpark** can process these streams and extract useful features such as posting frequency, number of interactions, repeated content, unusual login patterns, and account activity. Machine learning can then be used to identify suspicious accounts and misinformation. **Classification models such as Random Forest or Logistic Regression** can classify accounts as genuine or suspicious based on their behavior. For misinformation detection, text-analysis techniques can examine posts and identify suspicious or misleading content.

The system can store historical data in distributed cloud storage and use it to train and improve machine learning models. Real-time processing can compare new user activity with previously identified patterns and generate alerts when suspicious behavior is detected.

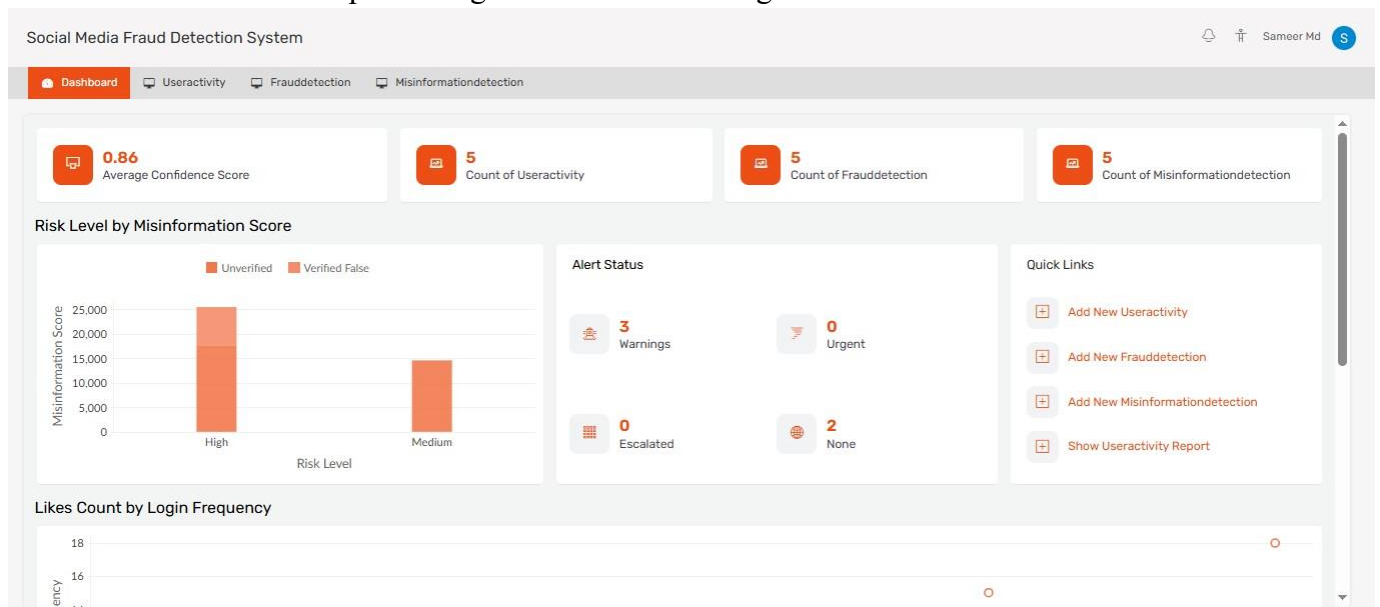
For security, the platform should use **authentication, access control, encryption, and monitoring** to protect user data. The solution should also be scalable so that additional processing resources can be added when the number of social-media interactions increases.

Architecture

Posts / Comments / Likes / Shares / User Activity



Thus, combining **Kafka for streaming**, **Spark for distributed processing**, and **machine learning for detection** provides a scalable solution for detecting fake accounts and misinformation with low processing delay. This directly addresses the assessment requirement for real-time/near-real-time processing and machine-learning-based fraud detection.



Social Media Fraud Detection System

Dashboard

Useractivity

Frauddetection

Misinformationdetection

Username

Activity Type

--Select--

Posts Count

Comments Count

Likes Count

Shares Count

Login Frequency

Activity Date

dd-MMM-yyyy

User Identifier

Session Id

Submit

Reset

Social Media Fraud Detection System

Dashboard

Useractivity

Frauddetection

Misinformationdetection

Useractivity Report

<

>

Edit

Duplicate

More

Username	Activity Type	Posts Count	Comments Count	Likes Count
eve_writer_33	Login	95	112	430
diana_gamer_77	Post	310	220	110
charlie_dev_x	Post	201	156	89
bob_trader_01	Comment	87	45	21
alice_fan_99	Share	142	89	52

Showing 5 of 5

Overview

Username

eve_writer_33

Activity Type

Login

Posts Count

95

Comments Count

112

Likes Count

430

Shares Count

28

Login Frequency

10

Activity Date

18-Aug-2026

User Identifier

uid_6610

Session Id

sess_7789e

Add a comment

Q3 – Government Data Platform

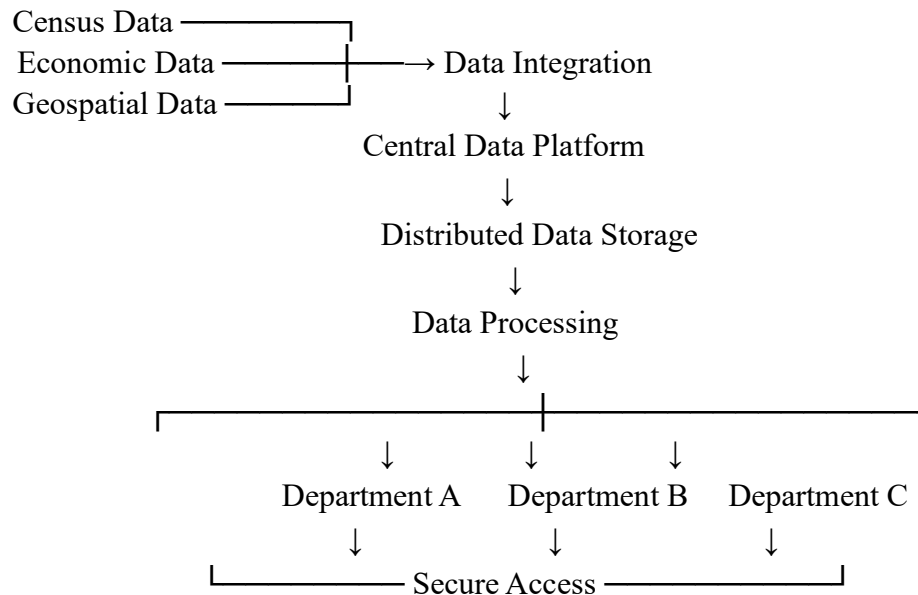
A government agency is developing a **national data platform** that combines large datasets such as **census data**, **economic data**, and **geospatial data**. Since different government departments need to access and share this information, the platform should provide a common and integrated data environment.

A **centralized Big Data platform** can be used to collect, store, integrate, and process data from multiple government agencies. Data can be stored in a scalable distributed storage system and processed using Big Data technologies. Standard data formats and APIs should be used to ensure **interoperability**, allowing different departments and systems to exchange and use data effectively.

A strong **data governance strategy** should define rules for data ownership, quality, access, privacy, security, and usage. Each department should have clearly defined access permissions based on its role. Sensitive citizen information should be protected using **encryption**, **authentication**, **role-based access control**, and **secure data transmission**.

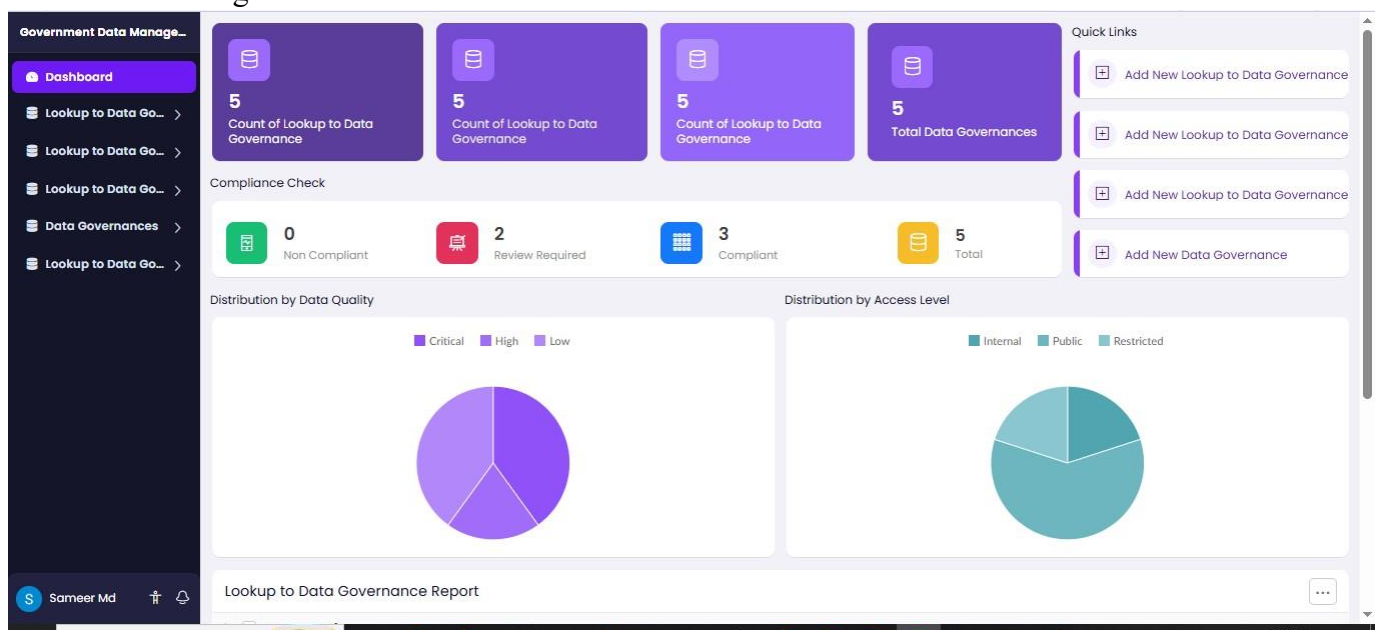
The platform should also maintain **data catalogs and metadata** so that departments can understand the source, format, meaning, and quality of datasets. Data quality checks should be performed before integrating information from different agencies.

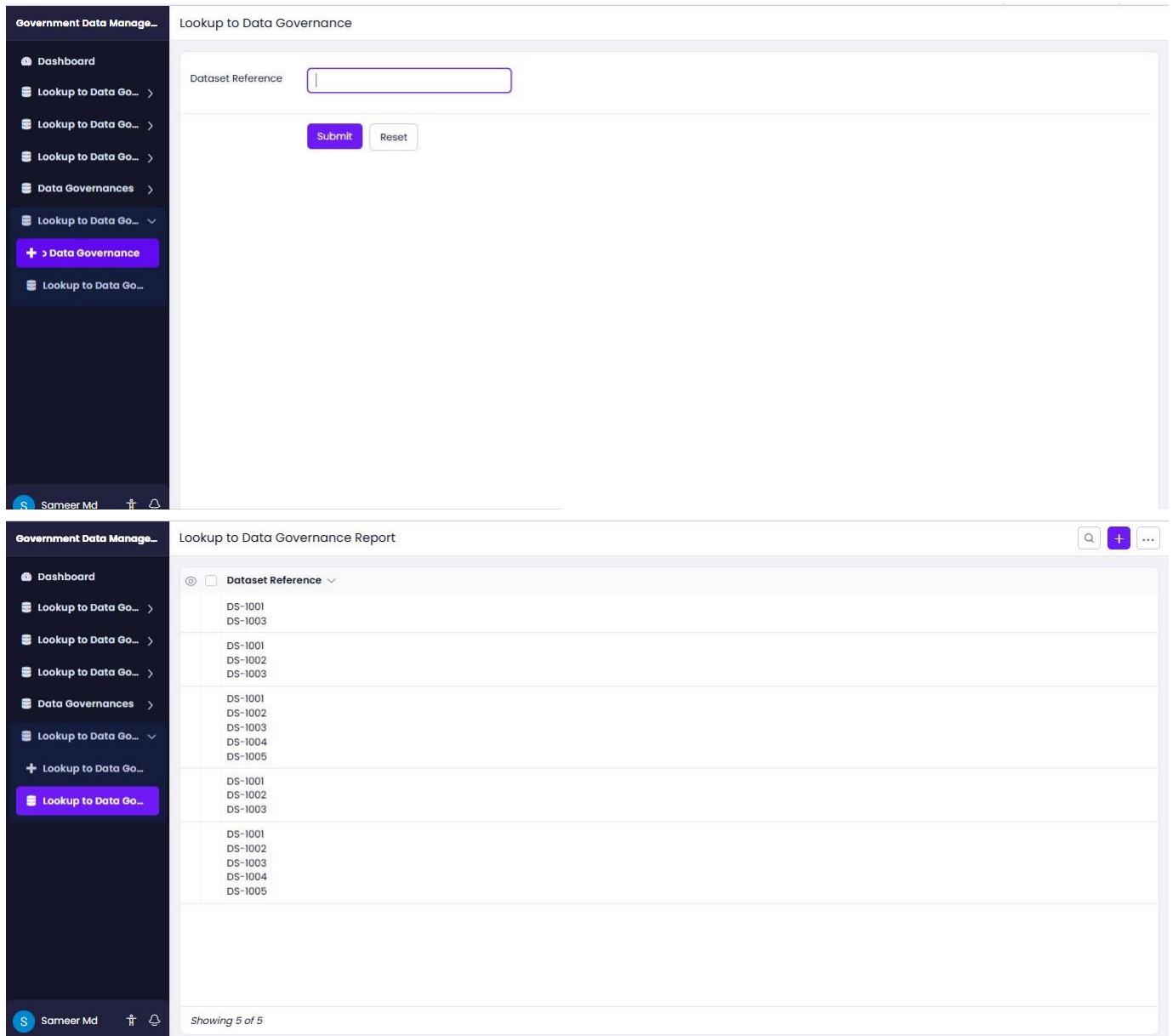
Proposed Architecture



The platform should implement **logging and monitoring** to track data access and detect unauthorized activities. Regular audits can help ensure that departments follow security and privacy policies.

Therefore, a centralized and governed Big Data platform can provide **interoperability, secure data sharing, data quality, privacy, and controlled access** across government agencies. This approach allows departments to use integrated census, economic, and geospatial information while maintaining strict security and privacy requirements. The solution directly addresses the assessment requirements for interoperability, privacy, security, governance, and integrated national data management.





Q4 – Banking Fraud Detection

A bank can implement a scalable **Big Data fraud detection pipeline** to analyze both historical transaction data and real-time transactions. Transaction information such as account number, transaction amount, location, time, device information, and transaction type can be collected continuously.

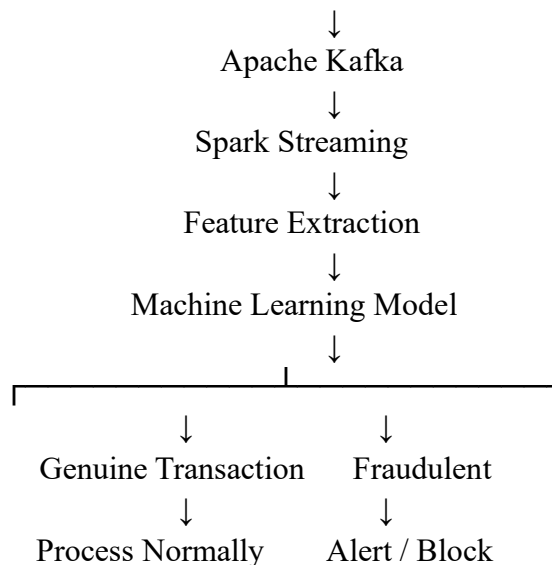
Apache Kafka can be used to receive and stream transactions in real time. **Apache Spark Streaming/PySpark** can process these transactions with low latency and extract important features such as unusual transaction amounts, transaction frequency, location changes, and abnormal spending patterns.

Historical transaction data can be stored in distributed cloud storage and used to train machine learning models. Algorithms such as **Random Forest, Logistic Regression, or anomaly-detection techniques** can be used to classify transactions as genuine or potentially fraudulent. The trained model can then analyze incoming transactions and generate a fraud score.

If a transaction is identified as highly suspicious, the system can generate an immediate alert for further verification. Continuous monitoring and model updates can improve detection accuracy over time.

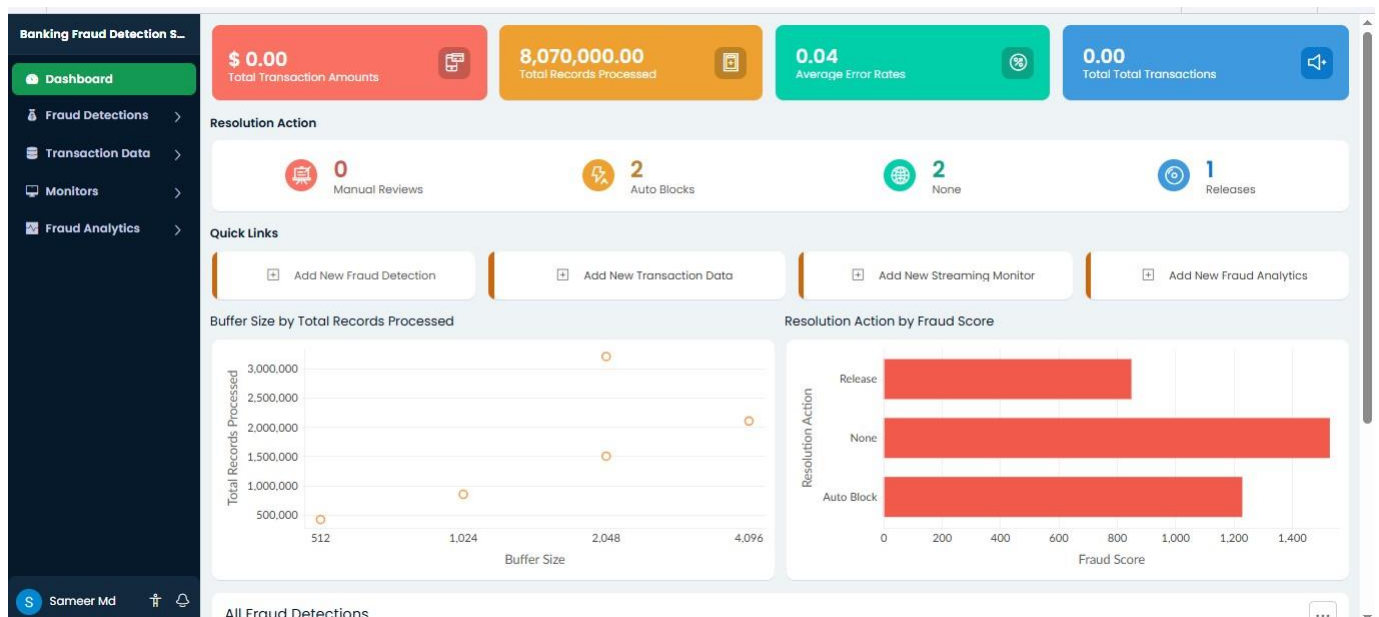
Big Data Pipeline

Historical Transactions + Real-Time Transactions



The system should also use **encryption, authentication, access control, logging, and monitoring** to protect sensitive financial information. A distributed architecture allows the system to scale when transaction volume increases.

Therefore, combining **Kafka for real-time streaming, Spark for distributed processing, and machine learning for fraud detection** provides a scalable, low-latency solution for detecting fraudulent banking transactions while maintaining high accuracy. This directly addresses the assessment requirements for historical and streaming data, low latency, high accuracy, technologies, and algorithms



Banking Fraud Detection S...

Dashboard

Fraud Detections

+ Fraud Detection

All Fraud Detections

Risk Levels

Transaction Data

Monitors

Fraud Analytics

Fraud Detection

Fraud Score

#####

Risk Level

-Select-

Detection Method

-Select-

Transaction Status

-Select-

Alert Status

-Select-

Detection Timestamp

dd-MMM-yyyy HH:mm:ss

Analyst Notes

Reviewed By

Review Date

dd-MMM-yyyy

Resolution Action

-Select-

Submit

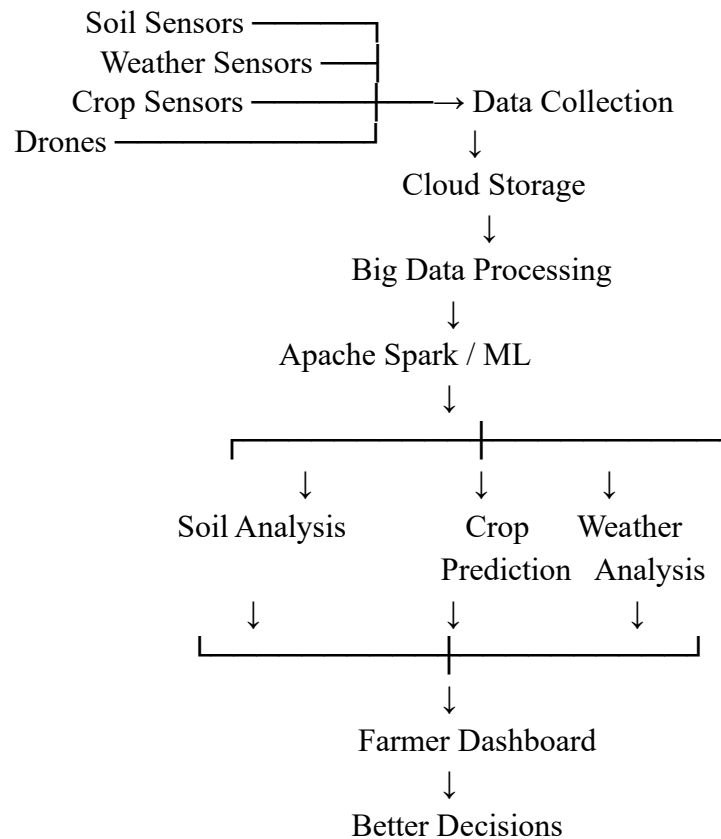
Reset

Q5 – Agriculture Smart Farming

Real-time sensor data can be processed using **stream-processing technologies**, while historical agricultural data can be processed using distributed Big Data tools such as **Apache Spark**. The collected data can include soil moisture, temperature, humidity, rainfall, crop health, and nutrient levels.

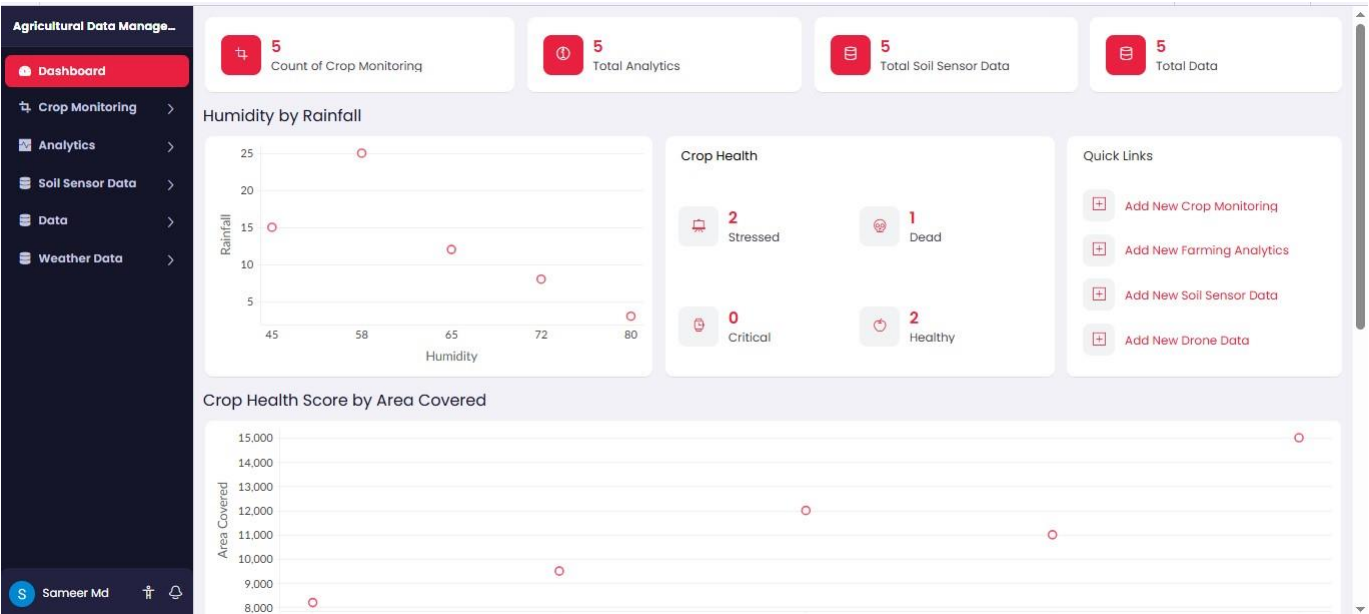
Drone images can also be analyzed to identify unhealthy crop areas. Based on these insights, farmers can make better decisions about **irrigation, fertilizer usage, pest control, and harvesting**. This can improve crop yield while reducing unnecessary use of water and other resources.

Big Data Architecture



The system should also provide **secure data access, authentication, monitoring, and controlled sharing** to protect agricultural data.

Therefore, Big Data analytics enables precision agriculture by combining distributed sensor and drone data to provide **real-time insights, crop predictions, optimized irrigation, efficient resource usage, and improved decision-making**. This directly addresses the assessment requirement for distributed agricultural data and analytics-based decision support.



Agricultural Data Manage...

Dashboard

Crop Monitoring

Analytics

Farming Analytics

All Analytics

Analytics Board

Soil Sensor Data

Data

Weather Data

S Sameer Md

Farming Analytics

Analytics Id

Field Code

Irrigation Recommendation

Fertilizer Recommendation

Disease Risk

Predicted Yield

Resource Usage

Recommendation Status

Soil Sensor Data

Weather Data

Submit

Reset

Agricultural Data Management System

Trial has expired

Upgrade

Edit this application

Help

Agricultural Data Manage...

Dashboard

Crop Monitoring

Analytics

Farming Analytics

All Analytics

Analytics Board

Soil Sensor Data

Data

Weather Data

Sameer Md

All Analytics

Analytics Id

Field Code

Irrigation

1005

Immedia

1004

Low

1003

None

1002

High

1001

High

Showing 5 of 5

Overview

Analytics Id	1004
Field Code	
Irrigation Recommendation	Low
Fertilizer Recommendation	Mixed
Disease Risk	560
Predicted Yield	61000
Resource Usage	1900
Recommendation Status	Pending
Soil Sensor Data	
Weather Data	

Weather Data

No records found

Field Code